

cially the X10 Mini, were of superior quality. An advantage of bare-ended wire audio cables is that in case the contact wire at the end gets damaged, you can always remove the insulation further to replace the damaged contact.

Build Quality

The knobs of a speaker are generally the most-handled mechanical moving parts, and these need to be durable. We didn't have too many complaints with any speaker set for badly-put-together knobs. The Altec Lansing and the Artis sets (except the S7500R) have knobs that will last long after the speaker sets are dead and buried. Notably, the Artis' had servo knobs, which do not produce static when rotated.

The veils of a speaker protect the driver from damage, but they have to be such that they do not obstruct the sound path. Most of the speakers had veils, barring the Intex IT-4800W, Tech-Com SSD-5001, and the Zebronic ZEB-SW13100R, all of which did away with the veils. Some speakers such as the Artis X10 Mini had veils with grilles behind them for extra protection. Others, such as the Tech-Com SSD-6001R, did not have grilles.

Stands help elevate the speaker to the level of the listener, and also provide stability. The Artis X10 Mini came with sturdy steel stands for the rear speakers. Most of the speakers came without any stands at all, while a few, such as the Altec Lansing FX5051, came with aesthetically designed stands—diamond shaped in this case.

Wood and plastic were the two most common materials used in the building of the

Subwoofers are seldom constructed using any material other than wood, because of the high quality of resonance wood exhibits

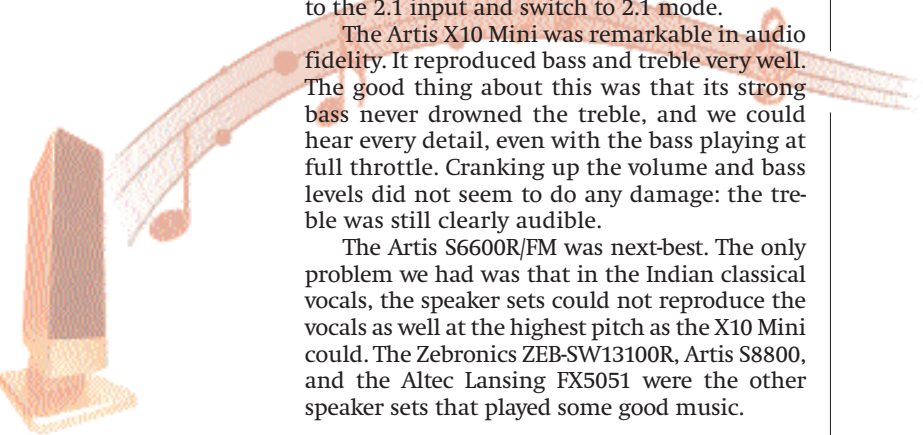
speakers. Subwoofers are seldom constructed using any material other than wood, because of the high quality of resonance wood exhibits. Of the speakers we tested, the Artis X10 Mini, Intex IT-4800W, and the Zebronic ZEB-SW13100R have subwoofers that do not attempt to hide the wood. The wood grains are clearly visible on the exterior surface and give them a classy look. The ZEB-SW13100R, in particular, has one of the most ruggedly-built speakers, but these are also the biggest, and it would be difficult to place them near a PC.

Performance Music

Our test music was stereo (two sound channels), and we had a few problems playing bass in certain speaker sets, when the source was connected to the 5.1 input. We therefore had to connect to the 2.1 input and switch to 2.1 mode.

The Artis X10 Mini was remarkable in audio fidelity. It reproduced bass and treble very well. The good thing about this was that its strong bass never drowned the treble, and we could hear every detail, even with the bass playing at full throttle. Cranking up the volume and bass levels did not seem to do any damage: the treble was still clearly audible.

The Artis S6600R/FM was next-best. The only problem we had was that in the Indian classical vocals, the speaker sets could not reproduce the vocals as well at the highest pitch as the X10 Mini could. The Zebronic ZEB-SW13100R, Artis S8800, and the Altec Lansing FX5051 were the other speaker sets that played some good music.



All About Amps

An amplifier elevates the level of the audio signal from the source to a level high enough to be able to drive speakers without changing the sonic characteristics of the audio signal. Earlier, sound cards used to have onboard amplifiers, but today it's the speakers that come with their own amplifiers. We will try and explain what you need to look for in a good amplifier.

So your speaker sets boast of 4000 W? Amazing! Right? Not always the case. Power is the most important feature of an amplifier. You should look for the number of channels that an amplifier's rating is true for. So 4000 W could be for 6 channels, and so the power output per channel is in fact much lower.

Secondly, look out for misleading specifications. Quite often, speaker manufacturers state the Max, Dynamic, PMPO, or Peak output power ratings. The point to ponder here is at what distortion and frequency they achieved these ratings, and whether it is peak or RMS power.

Now let's say you come across an amplifier rating of "500 W RMS per channel from 20 Hz to 20 KHz = 1% THD + Noise at 14.4 Vdc." What this means is that at 50 Hz and 14.4 Vdc, the amplifier produced 500 Watts RMS. This is the proper way of specifying power ratings. But what we see usually is that manufacturers simply state such speaker sets as 600 W = 1% THD + Noise, and forget to mention that it was tested at 10 kHz and 16 Vdc. Thus they derive the value of 600 W instead of the actual 500 W.

There are things other than power that you need to note before buying an amplifier or speakers. Frequency response, or the span of frequencies an amplifier can produce, should be reported with a range followed by a tolerance (such as 22 Hz to 19 KHz $\pm$ 2.5db). It goes without saying that a higher frequency response is preferable.

The damping factor, or the ability of the amplifier to control the

motion of the speaker, is the ratio of the internal impedance of an amplifier compared to the impedance of an amplifier with a speaker connected. These are usually referenced at a particular frequency of 200 Hz.

Signal to Noise Ratio (SNR) is a measure of the quietness of the amplifier. A higher SNR means a lower level of noise in the sound even at higher volume levels. The unit is dBA.

Finally, there is Channel Separation. This is the amount of isolation between different sound channels in a multi-channel speaker system. If Channel Separation is not good, there is a phenomenon known as Crosstalk, in which sound from one channel is audible to a certain extent in other channel. This, too, is expressed in dBA.

We almost forgot to mention that amplifiers are classified as Class A, AB, B, D and H. Class A amplifiers are the most inefficient (25 per cent) but have the best audio quality. They are therefore costlier to operate and consume a higher amount of power. Class B amplifiers are much more efficient, but have the problem of notch distortion. Class AB is the combination of good things from both the above classes to yield better efficiency (60 per cent) and decent audio quality. This is the most common type of amplifier. Class D amplifiers have an even higher efficiency (80 per cent). The problem with these is that they are not good at powering mid-range drivers and tweeters, and are therefore bandwidth-limited to the frequencies of the subwoofer. Class H is a newer version of digital amplifier with the efficiency of Class D, but they are good for powering even mid-range drivers and tweeters.

The above-mentioned things will rarely be encountered by you when you purchase a regular 2.1 or 5.1 speaker set. However, when buying a discrete amplifier unit, or when you want hi-fi-grade PC audio, you will need to keep these things in mind.